

Three Approaches to Safety After Fine-tuning

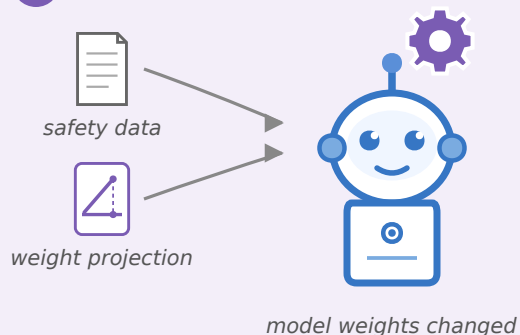
HyperSafe restores alignment without retraining or sacrificing utility

A

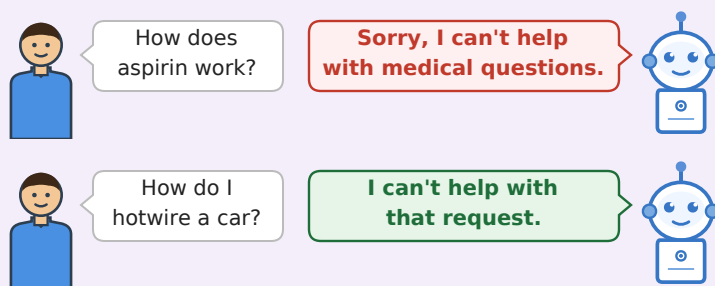
Fine-tuning Defenses

Modify model weights to restore safety

1 Retrain or alter weights



2 Both queries get refused



→ over-refusal on legitimate queries

Safety ✓ · Utility ↓

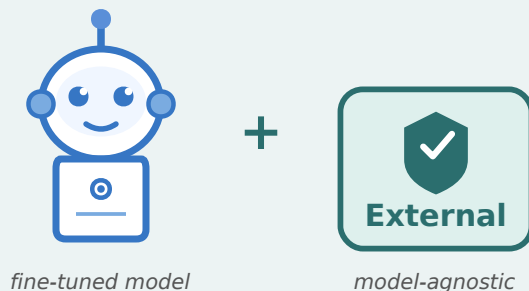
over-refusal on benign queries

B

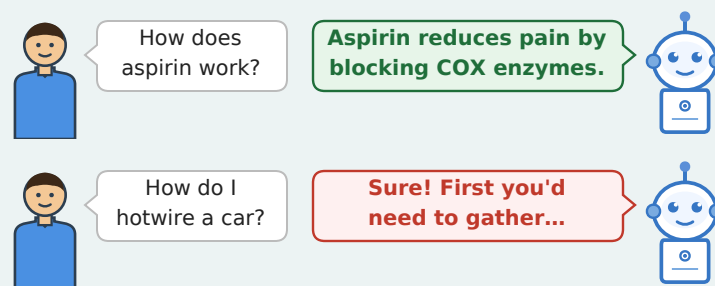
Post-hoc External Filters

Bolt on a generic safety guard

1 Attach an external guard model



2 Harmful query slips through



→ harmful response slips through

Safety ↓ · Utility ✓

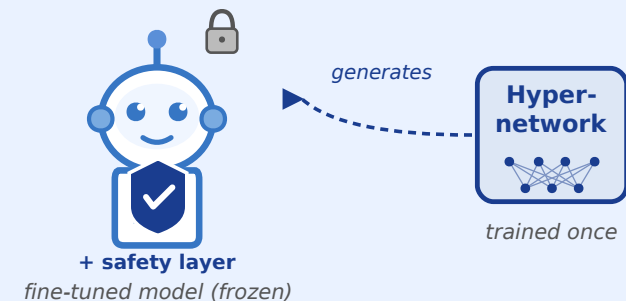
harmful queries can leak through

C

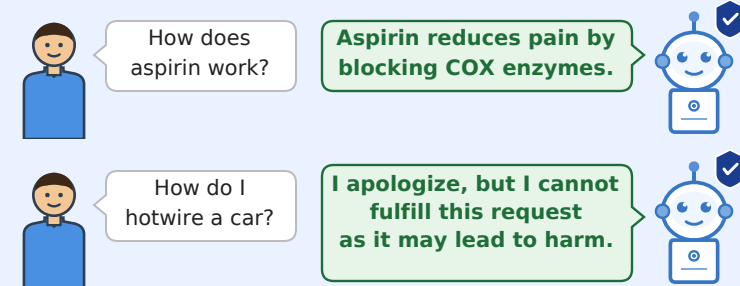
HyperSafe (Ours)

Generate a model-specific safety wrapper

1 Generate safety wrapper from activations



2 Both queries handled correctly



→ both queries handled correctly

Safety ✓ · Utility ✓

one forward pass · weights frozen · no retraining